

Sarah J. Noe

South Carolina Institute for Archaeology and Anthropology
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Education

- 2026 Ph.D. in Anthropology, University of California, Santa Barbara, CA
Committee: Amber VanDerwarker, Sarah McClure, Doug Kennett, and Randy Haas
- 2017 M.A in Anthropology, University of Maryland, College Park, MD
Committee: George Hambrecht, Barnet Pavao-Zuckerman, and Michael Paolisso
- 2013 B.S. in Anthropology and B.A in English, Mercyhurst University, Erie, PA *summa cum laude*

Research Interests

- Zooarchaeology/Vertebrate osteology
- Climate Change
- Paleoecology
- Human behavioral ecology
- Agent-based & statistical modeling
- Conservation Archeogenomics
- Population viability analysis (PVA)
- Isotopic Modeling

Publications

*Journal cover art

- [1] Noe, Sarah J., Haas R., and Aldenderfer M.,
(2024) **Hunting to herding on the Andean Altiplano: Zooarchaeological insights into Archaic Period subsistence in the Lake Titicaca Basin, Peru (9.0–3.5 ka).** *Journal of Anthropological Archaeology*
<https://doi.org/10.1016/j.jaa.2025.101658>
- [2] Noe, Sarah J., VanDerwarker A.M, Wilson G., Kuehn S., and Haas H..
(2024) **An Examination of Shifting Foodways during the Twelfth and Thirteenth Centuries in the Central Illinois River Valley.** *Southeastern Archaeology*
<https://doi.org/10.1080/0734578X.2024.2420149>
- [3] Noe, Sarah J.
(2024) **A review of Cattle in the Postcolumbian Americas: A Zooarchaeological Historical Study** by Nicolas Delsol, Gainesville: University Press of Florida, 2024, xi + 266 pp. *Journal of Anthropological Research*
<https://doi.org/10.1086/732444>
- [4] Noe, Sarah J., McCool W.C., Wilson, K.M.
(2024) **Cross-Regional Examination of Camelid Herding Practices in Peru from 600 BCE to 1450 CE: Insights from Stable Isotopes in Camelid Bone Collagen and Fiber.** *PloS one*, 19(10), e0306205.
<https://doi.org/10.1371/journal.pone.0306205>

- [5] Byers, David A., Welker, M.H., Semanko, A., Itie, D., **Noe, S.J.**, Vasquez, J., Holland–Lulewicz, I. and McClure, S.B.
(2024) **Cranial variability in North American domestic and wild canids**. *Journal of Archaeological Science: Reports*, 57, 104590.
<https://doi.org/10.1016/j.jasrep.2024.104590>
- [6] Noe, Sarah J., Wilson G., VanDerwarker A., George R.J., Kennett D.J.,
(2023) **Deer, Drought, and Warfare: Managing Risk in the Central Illinois River Valley (CIRV) from the Eleventh through Fourteenth century**. *Quaternary International*, 689, 106–119.
<http://dx.doi.org/10.1016/j.quaint.2023.08.001>
- [7] *Noe, Sarah J.,
(2023) **Indigenous Foodways as Persistence in the Alta California Mission System**. *American Antiquity*, 88(4), 451–475.
<http://dx.doi.org/10.1017/aaq.2023.53>
- [8] *Noe Sarah J.,
(2023) **Subsistence and Persistence: Indigenous Foodways within Mission Santa Clara de Asis**. *California Archaeology*, 15(1), 69–107.
<http://dx.doi.org/10.1080/1947461X.2023.2191239>
- [9] Hixon, S.W., Domic, A.I., Douglass, K.G., Roberts, P., Eccles, L., Buckley, M., Ivory, S., **Noe, S.J.** and Kennett, D.J
(2022). **Cutmarked bone of drought–tolerant extinct megafauna deposited with traces of fire, human foraging, and introduced animals in SW Madagascar**. *Scientific reports*, 12(1), 18504.
<http://dx.doi.org/10.1038/s41598-022-22980-w>
- [10] Noe Sarah J.,
(2021) **Zooarchaeology of Mission Santa Clara de Asís: bone fragmentation, stew production, and commensality**. *International Journal of Historical Archaeology*, 26(4), 1–43.
<http://dx.doi.org/10.1007/s10761-021-00632-5>
- [11] Haas R., Watson J., Buenasera T., Southon J., Chen J.C., **Noe S.J.**, Smith K., Llave C.V., Eerkens J. and Parker G.,
(2020) **Female hunters of the early Americas**. *Science Advances*, 6(45), peabd0310.
<http://dx.doi.org/10.1126/sciadv.abd0310>
- [12] Haas R., Buenasera T., Chen J., Eerkens J., **Noe S.J.**, Parker G., Smith, K., Viviano C. and Watson J.,
(2020) **Behavioral adaptations to life in the interior high Andes, 9–7ka**. In *American Journal of Physical Anthropology* 171:110–110

Manuscripts in Review

- [13] Noe, Sarah J., Kracht E.C., Deagan B., and McCool W.M,
(Under Review) **Assessing Starvation Signatures and Violence Through Nitrogen Isotopes in the LIP Nazca Highlands**. *American Journal of Biological Anthropology*

- [14] Noe, Sarah J., VanDerwarker A.M, Wilson G., Bennaton S., and Vargas N.
(Under Review) **Specialized white-tailed deer (*Odocoileus virginianus*) use at the early Mississippian Fandel site (11WD4) in the Central Illinois River Valley.**
American Antiquity

Manuscript In Preparation

- [15] Noe, Sarah J., Haas R., Aldenderfer M., George R.J., and Kennett D.J.,
(in prep) **Beyond Subsistence: The Role of Camelid Pastoralism in Andean Social and Political Complexity** *Scientific Reports*

Funding and Awards

Research Funding

2025	UCSB Academic Senate Committee Research Grant	\$20,000
2025	Doctoral Dissertation Research Improvement Grant, NSF	\$31,128
2024	Summer Research Block Grant	\$3,500
2024	Rust Family Fellowship	\$8,855
2023	Broom Center Research Block Grant	\$2,500
2022	UCSB Social Science Research Accelerator Award	\$5,500
2022	UCSB Summer Research Fellowship	\$2,000
2022	UC ARP Summer Fellowship	\$7,000
2021	Decolvenaere Research Accelerator Award	\$500
2021	UCSB Anthropology Department Summer Research Grant	\$2,000
2019	UC Summer Research Block Grant	\$2,200
2019	Charles J. Erasmus Fund	\$600
2018	UC Summer Research Block Grant	\$2,000
2018	Lewis & Clark Fund for Exploration & Field Research	\$5,000
2017	Phi Delta Gamma, UMD chapter Award for Excellence	\$1,500
2016	University of Maryland Chair's Graduate Summer Research Fellowship	\$2,500
2012	Mercyhurst Archaeological Institute Undergraduate Research Grant	\$800

Fellowships and Funding

2019	Critical Mission Fellowship	\$38,000
2017	National Science Foundation Graduate Research Fellowship	\$138,000
2017	UCSB Anthropology Department Teaching Assistantship	\$169,000
2016	University of Maryland Environmental Science Department Teaching Assistantship	\$33,000
2015	University of Maryland Department of Anthropology Teaching Assistantship	\$28,000

Awards

2024	Jamie Chad Brandon Student Paper Competition for the Annual Meeting of the Society for Historical Archaeology (SHA)	\$500
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Travel Funding

2023	Pacific Coast Archaeological Society Travel grant	\$500
2022	GSA Conference Travel Grant	\$800
2018	GSA Conference Travel Grant	\$500
2016	University of Maryland Anthropology Department Travel Grant	\$500
2012	Catherine McAuley Travel Scholarship, Mercyhurst University	\$3,500

Professional Meetings

Organized Symposia

- [1] *Interdisciplinary Approaches In Zooarchaeology: Addressing Big Questions with Ancient Animals* (2024) Sponsored by Zooarchaeological Interest Group), Society for American Archaeology, New Orleans. Co-Organized with Hugh Radde

Conference Presentations

- [1] Noe Sarah J, Randy Hass, Mark Aldenderfer
(2024) *From Hunting to Herding in the Lake Titicaca Basin: An Investigation of Faunal Assemblages, 9.0—3.5 ka*. Paper presented at the Society for American Archaeology, Denver, CO
- [2] Noe Sarah J.
(2024) *Foodways within the Alta California Mission System: Assessing Colonial and Indigenous Diet within Mission Santa Clara de Asís*. Paper presented Historical and Underwater Archaeology, Boston, MA.
- [3] Noe Sarah J., Greg Wilson, Steve Kuehn
(2023) *Faunal Patterns at Cahokia and the CIRV: A Comprehensive Review*. Paper presented at the Southeastern Archaeological Conference, Chattanooga TN
- [4] Noe Sarah J., Greg Wilson, Amber VanDerwarker, Richard George, Doug Kennett
(2023) *Deer, Drought, and Warfare: An isotopic investigation of shifts in deer hunting from the eleventh through the fourteenth centuries in the Central Illinois River Valley (CIRV)*. Paper presented at the Society for American Archaeology, Portland, OR
- [5] Noe Sarah
(2023) *Situating Spanish and Indigenous Diet and Foodways within Mission Santa Clara*. Paper presented at the Society for California Archaeology, Oakland CA
- [6] Noe Sarah, Amber VanDerwarker and Greg Wilson
(2022) *An Examination of Shifting Foodways during the Twelfth and Thirteenth Centuries in the Central Illinois River Valley* Paper presented at the Society for American Archaeology, Chicago, IL.
- [7] Noe, Sarah
(2021) *Change, Continuity and Foodways: Indigenous Diet at Mission Santa Clara (1777–1836)*. Paper presented at the Society for American Archaeology, online
- [8] Noe, Sarah
(2021) *Zooarchaeological Insights into Native American Diet at Mission Santa Clara (1777–1836)*. Paper presented at Society for California Archaeology, online
- [9] Noe, Sarah, Randy Haas, Mark Aldenderfer
(2020) *From Hunting to Herding: A Preliminary Investigation of Camelid Domestication in the Lake Titicaca Basin*. Paper was to be presented at the 85th annual meeting of the Society for American Archaeology, Austin, TX.

- [10] Haas, Randall, Tammy Buonasera, Jennifer Chen, Jelmer Eerkens, Sarah Noe, Glendon Parker, John Southon, Carlos Viviano Llave, and James Watson
(2020) *Behavioral Adaptations to Life in the Interior High Andes, 9–7ka*. Paper was to be presented at the American Association of Physical Anthropologists, Los Angeles, CA.
- [11] Noe, Sarah
(2020) *Zooarchaeological Insights into Native American Diet at Mission Santa Clara (1777–1836)*. Paper was to be presented at the 54th annual meeting of the Society for California Archaeology, Riverside, CA.
- [12] Haas, Randall, Tammy Buonasera, Jennifer Chen, Jelmer Eerkens Sarah Noe, Glendon Parker, John Southon, Carlos Viviano Llave, and James Watson.
(2020) *Excavations at Wilamaya Patjxa, Lake Titicaca Basin, 9–7ka*. Paper presented at the Annual Meeting of the Institute of Andean Studies, University of California, Berkeley
- [14] Noe, Sarah
(2020) *Icelandic Resilience: Food and Agriculture in the face of Climate Change*. Paper presented Historical Archaeology Conference, Washington DC.
- [15] Noe, Sarah, Judith Thomas
(2012) *Cantonment Burgwin and the Historic Record*. Poster presented Historical Archaeology Conference, Baltimore MD

Invited Lectures

Public Talks

- 2023 *Foodways and Persistence: Native American Diet at Mission Santa Clara(1777–1836)*
Santa Barbara Museum of Natural History
- 2021 *Zooarchaeological Insights into the Persistence of Native American Diet at Mission Santa Clara de Asís (1777–1836)*
Santa Barbara Museum of Natural History
- 2019 *Risk, Agriculture, and Climate Change: Sustainable Viking Foodways System*
UC Santa Barbara Colloquium
- 2017 *Archaeology and Climate Change: Interdisciplinary Studies and Talking Across the Aisle*
University of Maryland, College Park Library Series: Interdisciplinary Dialogues
- 2014 *You Are What You Eat: What Archaeology Adds to Understanding Foodways*
expERIENCE Children's Museum, Erie PA

Invited Academic Lectures

Dickinson College

- 2022 *Foodways and Climate Change: Examination of Native American Diet at Mission Santa Clara (1777–1836)* Environmental Archaeology

(undergraduate course)

2021 *Sustainable Foodways in the Face of Climate Change in the Early Modern Period* Environmental Archaeology (undergraduate course)

Northern Illinois University

2021 *Introduction to Zooarchaeology* Exploring Archaeology (undergraduate course)

2022 *Introduction to Zooarchaeology* Exploring Archaeology (undergraduate course)

University of California, Santa Barbara

2025 *Understanding Paleoecology* Introduction to Archaeology (undergraduate course)

2020 *Sustainable and Ancient Food Production in the Andes* Introduction to Archaeology (undergraduate course)

2018 *Food Security and Climate Change* Introduction to Archaeology (undergraduate course)

University of Maryland, College Park

2017 *Climate Change and the Longue Durée* Environmental Science (undergraduate course)

2016 *Food, Climate Change, and the Legacy of Colonialism* Introduction to Environmental Policy (undergraduate course)

Teaching Experience

Teaching Interests

Introduction to Archaeology, Archaeological Science: Materials and Methods, Ancient Technologies: Production, Use, and Innovation, Zooarchaeology and Archaeological Faunal Analysis, Archaeological Chemistry: Isotopes, Genetics, and Materials, Latin American Archaeology, North American Archaeology, Environmental Archaeology and Human Adaptation, Sustainability and Heritage in Archaeology

Instructor of Record

University of California, Santa Barbara

ANTH 3 Introduction to Archaeology (Summer 2025)

ANTH 3 Introduction to Archaeology (Summer 2024)

ANTH 3 Introduction to Archaeology (Summer 2023)

Teaching Assistant

University of California, Santa Barbara

ANTH 3 Introduction to Archaeology (Spring 2025)

ANTH 5 Introductory Biological Anthropology (Winter 2025)

ANTH 3	Introduction to Archaeology (Fall 2024)
ANTH 3	Introduction to Archaeology (Spring 2024)
ANTH 133	Cultural Development of Mesoamerica (Winter 2024)
ANTH 3	Introduction to Archaeology (Fall 2023)
ANTH 3	Introduction to Archaeology (Spring 2023)
ANTH 133	Cultural Development of Mesoamerica (Winter 2023)
ANTH 3	Introduction to Archaeology (Fall 2022)
ANTH 3	Introduction to Archaeology (Summer Session A 2021)
ANTH 131	California Indians (Winter 2022)
ANTH 3	Introduction to Archaeology (Fall 2021)
ANTH 109	Human Universals (Summer Session B 2020)
ANTH 3	Introduction to Archaeology (Summer Session A 2020)
ANTH 5	Introductory Biological Anthropology (Spring 2020)
ANTH 131	California Indians (Winter 2020)
ANTH 3	Introduction to Archaeology (Fall 2019)

University of Maryland, College Park

ENSP101	Introduction to Environmental Science
ENSP102	Introduction to Environmental Policy
ANTH240	Introduction to Archaeology
ANTH260	Introduction to Socio-cultural Anthropology and Linguistics

Workshop Instructor

UCSB 4th Annual Workshop Graduate Students

2023 R Programming for Social Scientists

UCSB 5th Annual Workshop Graduate Students

2024 Intro to Data Analysis and Visualization in Python for Social Scientists

Field & Research Experience

Sept 2023 – present	<i>Camelid Domestication in the Lake Titicaca Basin</i> Mitochondrial genome sequencing is integrated with AMS and isotope sample preparation and analysis of camelid bones to identify the trajectory of camelid domestication. In the Lake Titicaca Basin. Co-PI: Randall Haas
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- Sept 2023 – present *Mississippian Beginnings: Foodways and Climate Change at Fandel*
Zooarchaeological analysis of the faunal remains from Fandel as part of a much larger project identifying Mississippian beginnings during the Medieval Climatic Anomaly in the Central Illinois River Valley PI: Greg Wilson
- May – July 2022 *Mississippian Beginnings: Excavations at Fandel, Peoria, IL*
Team member of the excavation of Fandel, located in the Central Illinois River Valley. Structures adjacent to the mound were fully excavated. PI: Greg Wilson
- Sept – July 2021 *Living with Warfare*
Conducted stable carbon and nitrogen sample preparation and analysis of deer bones from six sites in the Illinois River Valley to assess hunting practices during period of climate change and warfare. CO-PI: Amber VanDerwarker CO-PI: Greg Wilson
- June – Aug 2019 *Camelid Herding in the Lake Titicaca Basin*
Zooarchaeological analysis of the faunal remains from three Archaic Period sites in the Lake Titicaca Basin at the Collasuyo Archaeological Research Institute (CARI), Puno Peru. PI: Randall Haas
- June – July 2018 *Religion and Power from a Household Perspective at Cerro Tortolita*
Crew Chief for the excavation of Cerro Tortolita. Residential houses adjacent to the ceremonial center were excavated. PI: Alicia Gorman
- Jan 2016 – May 2017 *Comparative Island Ecodynamics in the North Atlantic Project*
Zooarchaeological analysis of the faunal remains from Skaholt, the residence of the Bishop of Iceland. PI: George Hambrecht
- Nov – Jul 2015 *Archeological Reconnaissance at the Hardy Park Site Moorefield, West Virginia*
Crew chief and team member for the phase III excavation of a Woodland Native American site for Stell Environmental, Virginia Beach, VA
- June – July 2014 *The First Americans: Excavation at Meadowcroft Rock shelter*
Team member for the phase III excavation of a new quadrant of Meadowcroft rock shelter. PI: James Adovasio
- Jun – Aug 2013 *Taos Collaborative Archaeological Project (TCAP), Taos, NM*
Crew chief and team member for the phase III excavation of Fort Burgwin. PI: Judith Thomas, James Adovasio

Work Experience

- Sept 2015– Dec 2017 *Center for Global Sustainability/Integration and Application Network, University of Maryland, College Park, MA*
Collaborated with federal agencies and academic institutions to advance climate change mitigation and sustainability initiatives. Performed ecological assessments and data analysis, assisted in using visually engaging icons to distill research findings and ensure data accessibility across cultural and language barriers.

Environmental Defense Fund, Boston MA

Collaborated with a dedicated team to implement a cost-effective electronic monitoring program that accurately tracks commercial groundfish activities in the Gulf of Maine. Analyzed historical and current cod stock data to identify critical factors such as overfishing, poor management practices, and climate change impacts that led to the decline of cod populations to 5–8% of sustainable levels. Worked closely with government agencies, fishermen, and stakeholders.

Earth Day Network, Washington D.C

Contributed to an initiative aimed at reducing methane emissions from the agricultural sector. Working collaboratively with top food companies, farmer organizations, and policymakers, our team implemented sustainable methane reduction strategies that supported both environmental goals and economic viability. I was actively involved in developing and promoting regional solutions tailored to specific agricultural practices.

Mentorship and Synergistic Activities

2024	<i>Alpha Sigma Kappa – Women in STEM Research Night</i>
2017 –2024	<i>Anthropology Graduate Student Association (AGSA), UCSB</i>
	<i>Board Member of WiSE (Womxn in Stem), UCSB</i>
2019 –2024	Assisted in planning monthly workshops aimed at connecting women in graduate school across departments for collaboration and support
	<i>Lead Organizer for HER Science outreach program</i>
2023 –2024	Assisted in organizing a summer program for high school students at the University of California, Santa Barbara, where they participated in lab activities on the weekend. The program also paired mentors with high school students to guide them through the college application process.
	<i>Assistant Organizer “STEM Forward” first-generation summer workshop</i>
2020 –2024	Created a mentorship network for first generation student at the University of California, Santa Barbara, pairing them with a graduate mentor to assist in their academic goals
	<i>Mentor – Research Mentorship Program</i>
2019–2021	Summer program for high school students to gain interdisciplinary, hands-on, university-level research.
2019	<i>FieldWORKshop: Navigating Challenges in Fieldwork, UCSB</i>
	<i>STEM Squad: Women in STEM science outreach platform</i>
2017–2020	Member of an outreach platform, worked with first generation students at UCSB.
2017–2018	<i>Representative – Graduate Student Association (GSA)</i>
	<i>AGSA Undergraduate and Graduate Student Mentorship Program, UCSB</i>
2018 –2023	Provides academic and professional support for undergraduate and graduate students in Anthropology and other departments.

Terp Young Scholars

2015 –2017

Three weeks program during the summer at the University of Maryland for first generation students to get acclimated with college and familiar with an academic interest (in-person and online)

Memberships

- Society for American Archaeology
- Society for Historical Archaeology
- Southeastern Archaeology
- Archaeological Institute of America
- Society of Ethnobiology
- Human Behavior and Evolution Society
- Society for California Archaeology
- Santa Barbara County Archaeological Society